

## Effects of exercise training on older patients with major depression.

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**BACKGROUND:** Previous observational and interventional studies have suggested that regular physical exercise may be associated with reduced symptoms of depression. However, the extent to which exercise training may reduce depressive symptoms in older patients with major depressive disorder (MDD) has not been systematically evaluated. **OBJECTIVE:** To assess the effectiveness of an aerobic exercise program compared with standard medication (ie, antidepressants) for treatment of MDD in older patients, we conducted a 16-week randomized controlled trial. **METHODS:** One hundred fifty-six men and women with MDD (age,  $\geq 50$  years) were assigned randomly to a program of aerobic exercise, antidepressants (sertraline hydrochloride), or combined exercise and medication. Subjects underwent comprehensive evaluations of depression, including the presence and severity of MDD using Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition criteria and Hamilton Rating Scale for Depression (HAM-D) and Beck Depression Inventory (BDI) scores before and after treatment. Secondary outcome measures included aerobic capacity, life satisfaction, self-esteem, anxiety, and dysfunctional cognitions. **RESULTS:** After 16 weeks of treatment, the groups did not differ statistically on HAM-D or BDI scores ( $P = .67$ ); adjustment for baseline levels of depression yielded an essentially identical result. Growth curve models revealed that all groups exhibited statistically and clinically significant reductions on HAM-D and BDI scores. However, patients receiving medication alone exhibited the fastest initial response; among patients receiving combination therapy, those with less severe depressive symptoms initially showed a more rapid response than those with initially more severe depressive symptoms. **CONCLUSIONS:** An exercise training program may be considered an alternative to antidepressants for treatment of depression in older persons. Although antidepressants may facilitate a more rapid initial therapeutic response than exercise, after 16 weeks of treatment exercise was equally effective in reducing depression among patients with MDD.